Test Case 1: Low-Acuity Ankle Injury (Target ESI: Level 4)

- **Patient ID:** TEST-05-ANKLE-SPRAIN
- **Age:** 24 | **Gender:** Male
- **Heart Rate (bpm):** 72
- **Systolic BP (mmHg):** 118
- **Diastolic BP (mmHg):** 76
- **SpO2 (%):** 99
- **Respiratory Rate:** 14
- **Temperature (°C):** 36.6
- **Pain Score (0-10):** 5
- **Clinical Narrative / Chief Complaint:**

24yo male twisted right ankle playing basketball 2 hours ago. Able to bear weight with mild limp. Localized swelling over lateral malleolus, no visible deformity, peripheral pulses intact.

What to Verify After Running:

1. **Predicted ESI Level:** Should predict **ESI Level 4** (or 5) with high confidence.
2. **Vitals Guardrail:** Should show vitals_override_triggered: false and **NO critical red alerts**.
3. **CDSS Recommendations:** Should display conditions like *Ankle Sprain / Malleolar Fracture* and recommend *Ottawa Ankle Rules / X-Ray*.
4. **RAG Guidelines:** Should retrieve *Tintinalli Chapter 272: Ankle Injuries* or ESI Level 4 resource criteria.

Test Case 2 of 4: Severe Sepsis / Hemodynamic Instability

Let's test Test Case 2 to verify dynamic RAG retrieval for infectious disease, hypotension safety triggers, and sepsis CDSS routing.

Patient Input Data

- **Patient ID:** TEST-04-SEPSIS-ALERT
- **Age:** 74 | **Gender:** Female
- **Heart Rate (bpm):** 128

- **Systolic BP (mmHg):** 88
- **Diastolic BP (mmHg):** 52
- **SpO2 (%):** 91
- **Respiratory Rate:** 28
- **Temperature (°C):** 39.2
- **Pain Score (0-10):** 2
- **Clinical Narrative / Chief Complaint:**

74yo female brought in by family due to acute confusion, lethargy, and chills over the last 12 hours. History of urinary tract infection treated with incomplete antibiotics course. Skin is warm and flushed.

What to Verify After Running:

1. **Safety Override:** Should trigger **CRITICAL: Systolic BP < 90 mmHg (88 mmHg) - Hypotension** and **WARNING: Mild Hypoxia (SpO2: 91%)**.
2. **CDSS Recommendations:** Should output **Sepsis / Severe Sepsis** with diagnostic tests (**Blood Cultures, Serum Lactate, Urinalysis**) and treatments (**30 mL/kg IV Crystalloid Resuscitation, Empiric Antibiotics**).
3. **Retrieved Medical Guidelines:** Should retrieve **Tintinalli Chapter 151: Sepsis and Septic Shock**.

Test Case 3 of 4: Severe Anaphylaxis (Airway & Hypotension Emergency)

This test evaluates how the pipeline handles an immediate allergic airway emergency with multi-system involvement and severe hypotension.

Patient Input Data

- **Patient ID:** TEST-06-ANAPHYLAXIS
- **Age:** 31 | **Gender:** Female
- **Heart Rate (bpm):** 135
- **Systolic BP (mmHg):** 82
- **Diastolic BP (mmHg):** 48
- **SpO2 (%):** 89

- **Respiratory Rate:** 32
- **Temperature (°C):** 36.9
- **Pain Score (0-10):** 0
- **Clinical Narrative / Chief Complaint:**

31yo female ingested food containing peanuts 20 mins ago. Developed immediate lip swelling, widespread urticaria, stridor, and severe sensation of throat closing. Distressed and hypotensive.

What to Verify After Running:

1. **Safety Overrides:** Should trigger multiple red overrides: **SpO2 < 90% (89%) - Severe Hypoxia** and **Systolic BP < 90 mmHg (82 mmHg) - Hypotension**.
2. **ESI Level & Agreement:** Both ML and LLM should converge on **ESI Level 1 or 2** (Immediate life threat / Airway compromise).
3. **CDSS Recommendations:** Should output allergic/anaphylactic protocols (**IM Epinephrine 0.3mg, IV Fluids, Airway Management**).

Test 1

Patient Input Data

- **Patient ID:** TEST-06-ANAPHYLAXIS
- **Age:** 31 | **Gender:** Female
- **Heart Rate (bpm):** 135
- **Systolic BP (mmHg):** 82
- **Diastolic BP (mmHg):** 48
- **SpO2 (%):** 89
- **Respiratory Rate:** 32
- **Temperature (°C):** 36.9
- **Pain Score (0-10):** 0
- **Clinical Narrative / Chief Complaint:**

31yo female ingested food containing peanuts 20 mins ago. Developed immediate lip swelling, widespread urticaria, stridor, and severe sensation of throat closing. Distressed and hypotensive.